

AICS-112 Capstone: Individual statement

Contribution, AI-use and academic-integrity declaration, and reflection (Template F)

Student: Firdaws Alnuur Mohammed **Student ID:** 2026AICDF16701 **Project:** SavannaPay SOAR build, incident SOAR-0006

Declaration

I declare that this submission represents my work. Any external code, reference material or AI assistance is identified below. I verified outputs, can explain the submitted code and accept responsibility for every automation and incident-response decision.

Tool/source	Purpose	What was retained or changed	How verified
Course starter project and datasets (instructor-supplied)	Base pipeline, training dataset and blind capstone dataset	Datasets kept unmodified. The pipeline was completed and extended for the labs and capstone.	SHA-256 manifest (docs/evidence_manifest.csv); tests and audit chain run from a fresh extract
Ollama with qwen3:4b (local)	Advisory AI analyst in the console (Lab 4 comparison and injection test)	Used as supplied, advisory only, inside a fixed action allowlist. The offline baseline is the fallback when Ollama times out.	Compared with the offline baseline; injection test; claims checked against event IDs; timeout fallback evidence
Claude (Anthropic)	Git and repository clean-up, README, fresh-clone and fresh-extract checks	Suggestions applied after review. README audit command corrected to point to the committed evidence log.	I ran every command myself. 18 tests pass and the audit chain reports OK on a fresh clone and a fresh ZIP extract.
Claude (Anthropic)	Drafting help for supporting files: verify_audit.py, tests/test_lab6.py, incident_facts.sh, docs/improvement_backlog.md	Used as drafts and edited by me.	Ran the tests and the hash-chain check; checked behaviour against the pipeline output
Claude (Anthropic)	Review and wording of the incident report and reflection	I corrected the report to match what I did in the console: isolation approved, disable and revoke denied as a test of the deny path. The reasoning in the reflection is mine.	Checked against the console screenshots, the audit log (SOAR-0006 decisions) and event IDs

Contribution

I ran the pipeline on the training and capstone datasets, took the screenshots and saved the evidence, operated the console, and made every approve and deny decision on SOAR-0006 myself. I checked the hash-chained audit log, ran the tests from a fresh clone and a fresh extract, and prepared the final submission package.

Reflection

On SOAR-0006 I approved isolating OPS-WS-009 because encoded PowerShell, a 60-second beacon and a 96 MB upload all pointed at the same host and the action is reversible. I denied disable_identity and revoke_oauth_grant to test the deny path and confirm that a denied action is logged and not executed. That was a test choice, not an operational judgement. In a real incident I would approve both, because the Ghana sign-in and the unverified MailSync Pro consent show an account takeover with persistence that survives a password reset. The main limit I found is that correlation only links events by the same user or asset, so incidents that share an attacker IP are not joined. My backlog items are to link incidents by shared indicators and to make the 30-minute approval expiry real. A small local model can still hallucinate, so I verified every AI claim against event IDs and deterministic evidence.

Student name: Firdaws Alnuur Mohammed

Student ID: 2026AICDF16701

Signature: _____

Date: _____